
Parte 2: Procesamiento de consultas

1. Resultado consulta dos tras la creación del índice sobre nacionalidad

```
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'Actors.Voice-actors',
    parsedQuery: {
      '$and': [
        {
          Idiomas: {
            '$eq': 'Ingles'
          }
        },
        {
          Nacionalidad: {
            '$eq': 'Chile'
          }
        }
      ]
    },
    indexFilterSet: false,
    queryHash: 'F3BD70B4',
    planCacheShapeHash: 'F3BD70B4',
    planCacheKey: '5DDE1143',
    optimizationTimeMillis: 25,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'PROJECTION_SIMPLE',
      transformBy: {
        _id: 0,
        Nombre: 1,
        Nacionalidad: 1,
        Idiomas: 1
      },
      inputStage: {
        stage: 'SORT',
        sortPattern: {
          Nombre: 1
        },
        memLimit: 104857600,
        limitAmount: 10,
        type: 'simple',
        inputStage: {
          stage: 'FETCH',
          filter: {
            Idiomas: {
              '$eq': 'Ingles'
            }
          },
          inputStage: {
```

```

        stage: 'IXSCAN',
        keyPattern: {
          Nacionalidad: 1
        },
        indexName: 'Nacionalidad_1',
        isMultiKey: false,
        multiKeyPaths: {
          Nacionalidad: []
        },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: {
          Nacionalidad: [
            '["Chile", "Chile"]'
          ]
        }
      }
    }
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 6,
  executionTimeMillis: 33,
  totalKeysExamined: 6,
  totalDocsExamined: 6,
  executionStages: {
    isCached: false,
    stage: 'PROJECTION_SIMPLE',
    nReturned: 6,
    executionTimeMillisEstimate: 0,
    works: 14,
    advanced: 6,
    needTime: 7,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    transformBy: {
      _id: 0,
      Nombre: 1,
      Nacionalidad: 1,
      Idiomas: 1
    }
  },
  inputStage: {
    stage: 'SORT',
    nReturned: 6,
    executionTimeMillisEstimate: 0,
    works: 14,
    advanced: 6,
    needTime: 7,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    sortPattern: {
      Nombre: 1
    }
  }
}

```

```

memLimit: 104857600,
limitAmount: 10,
type: 'simple',
totalDataSizeSorted: 3702,
usedDisk: false,
spills: 0,
spilledDataStorageSize: 0,
inputStage: {
  stage: 'FETCH',
  filter: {
    Idioms: {
      '$eq': 'Ingles'
    }
  },
  nReturned: 6,
  executionTimeMillisEstimate: 0,
  works: 7,
  advanced: 6,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  docsExamined: 6,
  alreadyHasObj: 0,
  inputStage: {
    stage: 'IXSCAN',
    nReturned: 6,
    executionTimeMillisEstimate: 0,
    works: 7,
    advanced: 6,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    keyPattern: {
      Nacionalidad: 1
    },
    indexName: 'Nacionalidad_1',
    isMultiKey: false,
    multiKeyPaths: {
      Nacionalidad: []
    },
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'forward',
    indexBounds: {
      Nacionalidad: [
        '["Chile", "Chile"]'
      ]
    },
    keysExamined: 6,
    seeks: 1,
    dupsTested: 0,
    dupsDropped: 0
  }
},
},
},
},
},

```

```

queryShapeHash: 'FBD5590A7B8E5C87163EAA81455789FD2C4973D8175382668E2C98FBFD3C289F',
command: {
  find: 'Voice-actors',
  filter: {
    Nacionalidad: 'Chile',
    Idiomas: 'Ingles'
  },
  sort: {
    Nombre: 1
  },
  projection: {
    _id: 0,
    Nombre: 1,
    Nacionalidad: 1,
    Idiomas: 1
  },
  limit: 10,
  '$db': 'Actors'
},
serverInfo: {
  host: 'cecea55db233',
  port: 30002,
  version: '8.0.29',
  gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1788134576, i: 1 }),
  signature: {
    hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
    keyId: Long('0')
  }
},
operationTime: Timestamp({ t: 1788134576, i: 1 })
}

```

2. Resultado consulta uno tras la creación del índice sobre edad

```
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'Actors.Voice-actors',
    parsedQuery: {
      Edad: {
        '$gt': 50
      }
    },
    indexFilterSet: false,
    queryHash: '07B623A2',
    planCacheShapeHash: '07B623A2',
    planCacheKey: 'E68DE247',
    optimizationTimeMillis: 584,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'LIMIT',
      limitAmount: 10,
      inputStage: {
        stage: 'PROJECTION_SIMPLE',
        transformBy: {
          _id: 0,
          Nombre: 1,
          Edad: 1,
          Nacionalidad: 1
        }
      },
      inputStage: {
        stage: 'FETCH',
        inputStage: {
          stage: 'IXSCAN',
          keyPattern: {
            Edad: 1
          },
          indexName: 'Edad_1',
          isMultiKey: false,
          multiKeyPaths: {
            Edad: []
          },
          isUnique: false,
          isSparse: false,
          isPartial: false,
          indexVersion: 2,
          direction: 'forward',
          indexBounds: {
            Edad: [
              '(50, inf.0]'
            ]
          }
        }
      }
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
```

```
nReturned: 7,
executionTimeMillis: 815,
totalKeysExamined: 7,
totalDocsExamined: 7,
executionStages: {
  isCached: false,
  stage: 'LIMIT',
  nReturned: 7,
  executionTimeMillisEstimate: 187,
  works: 8,
  advanced: 7,
  needTime: 0,
  needYield: 0,
  saveState: 2,
  restoreState: 2,
  isEOF: 1,
  limitAmount: 10,
  inputStage: {
    stage: 'PROJECTION_SIMPLE',
    nReturned: 7,
    executionTimeMillisEstimate: 187,
    works: 8,
    advanced: 7,
    needTime: 0,
    needYield: 0,
    saveState: 2,
    restoreState: 2,
    isEOF: 1,
    transformBy: {
      _id: 0,
      Nombre: 1,
      Edad: 1,
      Nacionalidad: 1
    },
    inputStage: {
      stage: 'FETCH',
      nReturned: 7,
      executionTimeMillisEstimate: 187,
      works: 8,
      advanced: 7,
      needTime: 0,
      needYield: 0,
      saveState: 2,
      restoreState: 2,
      isEOF: 1,
      docsExamined: 7,
      alreadyHasObj: 0,
      inputStage: {
        stage: 'IXSCAN',
        nReturned: 7,
        executionTimeMillisEstimate: 177,
        works: 8,
        advanced: 7,
        needTime: 0,
        needYield: 0,
        saveState: 2,
        restoreState: 2,
        isEOF: 1,
        keyPattern: {
          Edad: 1
        },
        indexName: 'Edad_1',
        isMultiKey: false,
```

```

        multiKeyPaths: {
            Edad: []
        },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: {
            Edad: [
                '(50, inf.0]'
            ]
        },
        keysExamined: 7,
        seeks: 1,
        dupsTested: 0,
        dupsDropped: 0
    }
}
}
},
queryShapeHash: '21893A6C9EF92D4FCE3655C3939BC0D9115EAB01062C13D8FD87776F9FDFF6A3 ',
command: {
    find: 'Voice-actors',
    filter: {
        Edad: {
            '$gt': 50
        }
    },
    sort: {
        Edad: 1
    },
    projection: {
        _id: 0,
        Nombre: 1,
        Edad: 1,
        Nacionalidad: 1
    },
    limit: 10,
    '$db': 'Actors'
},
serverInfo: {
    host: 'cecea55db233',
    port: 30002,
    version: '8.0.29',
    gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148 '
},
serverParameters: {
    internalQueryFacetBufferSizeBytes: 104857600,
    internalQueryFacetMaxOutputDocSizeBytes: 104857600,
    internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
    internalDocumentSourceGroupMaxMemoryBytes: 104857600,
    internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
    internalQueryProhibitBlockingMergeOnMongoS: 0,
    internalQueryMaxAddToSetBytes: 104857600,
    internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
    internalQueryFrameworkControl: 'trySbeRestricted',
    internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
    clusterTime: Timestamp({ t: 1788128321, i: 1 }),

```

```

signature: {
  hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
  keyId: Long('0')
},
operationTime: Timestamp({ t: 1788128321, i: 1 })
}

```

3. Resultado consulta tres tras la creación del índice sobre edad

```

{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'Actors.Voice-actors',
    parsedQuery: {
      '$and': [
        {
          Personajes: {
            '$elemMatch': {
              '$and': [
                {
                  Generos: {
                    '$eq': 'Aventura'
                  }
                },
                {
                  Rol: {
                    '$eq': 'principal'
                  }
                }
              ]
            }
          }
        ]
      ],
      Edad: {
        '$lt': 30
      }
    ]
  },
  indexFilterSet: false,
  queryHash: 'E4BD0639',
  planCacheShapeHash: 'E4BD0639',
  planCacheKey: '74AB3BFB',
  optimizationTimeMillis: 5,
  maxIndexedOrSolutionsReached: false,
  maxIndexedAndSolutionsReached: false,
  maxScansToExplodeReached: false,
  prunedSimilarIndexes: false,
  winningPlan: {
    isCached: false,
    stage: 'LIMIT',
    limitAmount: 10,
    inputStage: {
      stage: 'PROJECTION_DEFAULT',
      transformBy: {
        _id: 0,
        Nombre: 1,

```

```

    Edad: 1,
    'Personajes.Rol': 1,
    'Personajes.Generos': 1
  },
  inputStage: {
    stage: 'FETCH',
    filter: {
      Personajes: {
        '$elemMatch': {
          '$and': [
            {
              Generos: {
                '$eq': 'Aventura'
              }
            },
            {
              Rol: {
                '$eq': 'principal'
              }
            }
          ]
        }
      }
    },
    inputStage: {
      stage: 'IXSCAN',
      keyPattern: {
        Edad: 1
      },
      indexName: 'Edad_1',
      isMultiKey: false,
      multiKeyPaths: {
        Edad: []
      },
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'forward',
      indexBounds: {
        Edad: [
          '[-inf.0, 30)'
        ]
      }
    }
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 4,
  executionTimeMillis: 6,
  totalKeysExamined: 6,
  totalDocsExamined: 6,
  executionStages: {
    isCached: false,
    stage: 'LIMIT',
    nReturned: 4,
    executionTimeMillisEstimate: 0,
    works: 7,
    advanced: 4,
  }
}

```

```

needTime: 2,
needYield: 0,
saveState: 0,
restoreState: 0,
isEOF: 1,
limitAmount: 10,
inputStage: {
  stage: 'PROJECTION_DEFAULT',
  nReturned: 4,
  executionTimeMillisEstimate: 0,
  works: 7,
  advanced: 4,
  needTime: 2,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  transformBy: {
    _id: 0,
    Nombre: 1,
    Edad: 1,
    'Personajes.Rol': 1,
    'Personajes.Generos': 1
  },
  inputStage: {
    stage: 'FETCH',
    filter: {
      Personajes: {
        '$elemMatch': {
          '$and': [
            {
              Generos: {
                '$eq': 'Aventura'
              }
            },
            {
              Rol: {
                '$eq': 'principal'
              }
            }
          ]
        }
      }
    }
  },
  nReturned: 4,
  executionTimeMillisEstimate: 0,
  works: 7,
  advanced: 4,
  needTime: 2,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  docsExamined: 6,
  alreadyHasObj: 0,
  inputStage: {
    stage: 'IXSCAN',
    nReturned: 6,
    executionTimeMillisEstimate: 0,
    works: 7,
    advanced: 6,
    needTime: 0,
    needYield: 0,

```

```

        saveState: 0,
        restoreState: 0,
        isEOF: 1,
        keyPattern: {
            Edad: 1
        },
        indexName: 'Edad_1',
        isMultiKey: false,
        multiKeyPaths: {
            Edad: []
        },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: {
            Edad: [
                '[-inf.0, 30)'
            ]
        },
        keysExamined: 6,
        seeks: 1,
        dupsTested: 0,
        dupsDropped: 0
    }
}
}
}
},
queryShapeHash: '74B9733C635319C8A04021D3B8412F204CDBBF8F31453A13449D532A3877E557 ',
command: {
    find: 'Voice-actors',
    filter: {
        Edad: {
            '$lt': 30
        },
        Personajes: {
            '$elemMatch': {
                Rol: 'principal',
                Generos: 'Aventura'
            }
        }
    },
    sort: {
        Edad: 1
    },
    projection: {
        _id: 0,
        Nombre: 1,
        Edad: 1,
        'Personajes.Rol': 1,
        'Personajes.Generos': 1
    },
    limit: 10,
    '$db': 'Actors'
},
serverInfo: {
    host: 'cecea55db233',
    port: 30002,
    version: '8.0.29',
    gitVersion: '559d67c651f7393e62757234a0b25cb7a8622148 '
},

```

```
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1,
'$clusterTime': {
  clusterTime: Timestamp({ t: 1788136720, i: 1 }),
  signature: {
    hash: Binary.createFromBase64('AAAAAAAAAAAAAAAAAAAAAAAAAAAA=', 0),
    keyId: Long('0')
  }
},
operationTime: Timestamp({ t: 1788136720, i: 1 })
}
```